

Build a Static Website

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Purpose

In this lab, I made a static website by creating a bucket, adding objects to it (in other words the websites), and finally making bucket policies. This teaches me other elements of using Amazon AWS and more about VPCs, specifically showing me how to create a static website through them.

Lab Walkthrough (with Documentation)

Step 1: Create A Bucket

General configuration

Bucket name

Bucket name must be unique and must not contain spaces or uppercase letters. [See rules for bucket naming](#)

AWS Region

US East (N. Virginia) us-east-1

Copy settings from existing bucket - *optional*
Only the bucket settings in the following configuration are copied.

Choose bucket

Step 2: Enable Website Hosting

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

☐ Disable

☒ Enable

Hosting type

☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)

☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document

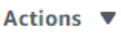
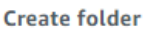
Specify the home or default page of the website.

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Step 3: Upload Website Files (Objects)

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

< 1 > 

☐	Name ▲	Type ▼	Last modified ▼	Size ▼	Storage class ▼
☐	 bitbangers.png	png	March 19, 2021, 21:28:00 (UTC-07:00)	56.5 KB	Standard
☐	 index.html	html	March 19, 2021, 19:27:40 (UTC-07:00)	277.0 B	Standard

Step 4: Allow Public Access

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block *all* public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☐ **Block public access to buckets and objects granted through *new* access control lists (ACLs)**

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☐ **Block public access to buckets and objects granted through *any* access control lists (ACLs)**

S3 will ignore all ACLs that grant public access to buckets and objects.
- ☐ **Block public access to buckets and objects granted through *new* public bucket or access point policies**

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☐ **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

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Step 5: Confirm Unblock Changes

Turning this setting on is the same as turning on all four settings below. Each of the following settings are

Edit Block public access (bucket settings) ×

 This will result in public access being blocked for this bucket and all objects in the bucket.

To confirm the settings, enter *confirm* in the field.

Cancel Confirm

Step 6: Add Bucket Policy

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#) 

Policy examples  Policy generator 

Bucket ARN

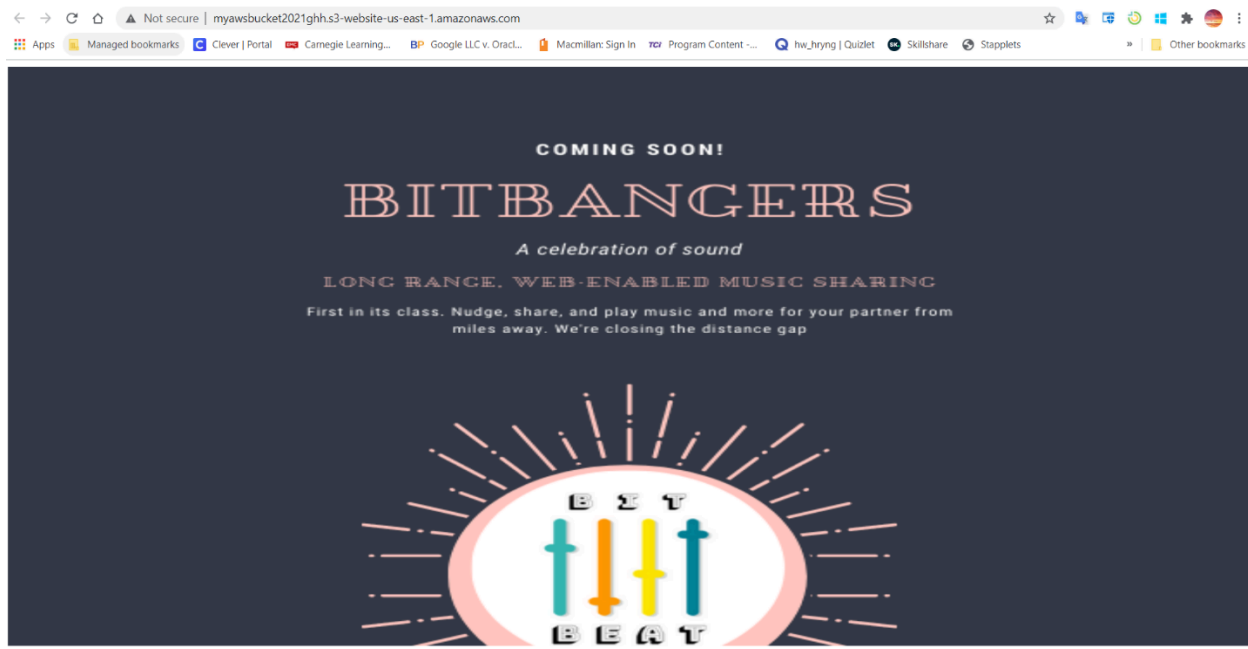
 `arn:aws:s3:::myawsbucket2021ghh`

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [{
4     "Sid": "PublicReadGetObject",
5     "Effect": "Allow",
6     "Principal": "*",
7     "Action": ["s3:GetObject"],
8     "Resource": ["arn:aws:s3:::myawsbucket2021ghh/*"]
9   }]
10 }
11 ]
12 }
```

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Step 7: Confirm Static Website Works



Problems

One issue I had during this lab was when I was copying the policy given to me in the lab instructions, there was a line in the policy that had an error and thus I could not add this policy. I clicked over the 'X' to see what the problem was, and it expected something besides the letter that started the line. When I got rid of it, the error was still there. After trying a few different things, I decided instead of copying the policy given and pasting to type each line one by one. After this the errors went away, and the formatting looked correct.

However, when I tried to add the policy, an error message told me I did not have permission to add or edit policies. I found out I had misread the instructions and kept public access off instead of unblocking. After going back to do so, I was able to edit policies.

Lastly, I followed all the instructions, and at the end it said to confirm that my website is accessible. I copied the link but changed the bucket name to the one I created. When trying to go to the page, I got an error message saying that either my link was wrong or the page does not exist. I tried again and typed the link out this time carefully. Unfortunately, that did not work either. I talked to some of my peers, sharing my screenshots to have another set of eyes double-check my work and they said it appeared to be correct. After some time, my peer found that I did not change the region to the correct region in the URL. I needed it to be "us-east-1," however even after adding that, the link did not work. My peer later remembered that at the bottom of the Bucket Properties tab, there is a link you can click to reach the website. Through this, I was able to access the static webpage. I found that when I had changed the sample URL "website-region" part, that I removed both of those words when I needed to keep in the word "website" and only change the "region" part.

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